

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper No. STB-401: Sample Surveys and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

BS | Sem -W] 132, Note: Answer any FIVE questions. All questions carry equal marks.

Question 1.

- (a) Distinguish between sampling and complete enumeration with examples. What steps should be followed in a sample survey.
- (b) A population consists of five units 10, 25, 30, 35 and 45, Give all possible samples of size 2 that can be drawn using SRSWR and show that the sample mean is an unbiased estimator of population mean.

Question 2.

- (a) Prove that the probability of selecting a particular unit of the population is same for each draw in SRSWOR scheme. Also show that the sample mean square is an unbiased estimator of population mean square in SRSWOR.
- (b) Describe simple random sampling of attribute. Find out the sampling variance of the unbiased estimator of population proportion in SRSWOR.

Question 3.

- (a) Define an unbiased estimator of population mean in stratified random sampling and obtain its sampling variance under proportional allocation.
- (b) Explain the purpose of stratification in sample surveys. Show that the estimator of population mean under proportional allocation provides better estimate than simple random sampling.

Question 4.

- (a) Describe systematic sampling and its utility. Compare the efficiency of systematic sampling with that of simple random sampling.
- (b) What is cluster sampling? Compare the relative efficiency of cluster sampling with that of simple random sampling.

Question 5.

- (a) Derive the expression of the situation for which ratio estimator is preferred over the sample mean estimator. If coefficients of variation of study and auxiliary variables are $C_y = 0.02$ and $C_x = 0.01$ respectively. Suggest the appropriate estimator for estimating the population mean of study variable, where the correlation coefficient between study and auxiliary variables is $\rho = 0.8$.
 - (b) Describe the regression method of estimation. Find out the mean square error of the regression estimator of population mean and compare it with that of product estimator.
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Question 6.

- (a) What is analysis of variance? Describe it with a suitable example.
 - (b) Work out the analysis of variance for two-way classified data with one observation per cell, giving its model and complete analysis.
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Question 7.

- (a) Explain the purpose of design of experiments. Describe the following fundamental Principles:
 - (b) Randomization, (i) Replication and (iii) Local control.
 - (c) Give layout and complete statistical analysis of completely randomized design. What are the advantages and disadvantages of this design?
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Question 8. Write short notes on any three of the following:

- (a) Basic principles of sample survey
 - (b) Sampling and non-sampling errors
 - (c) Optimum allocation
 - (d) Product method of estimation
 - (e) Latin square design
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